

Obligatory Exercise 2

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Exercise O2.1

$$F_1 : ((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

To prove the validity of F_1 in the resolution calculus we can prove that it is not the opposite of that instead. That would be by proving unsatisfiability of negated form of F_1 instead which becomes $\neg(((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r))$. Next we must transform this formula into a clausal form.

$$\begin{aligned}\neg F_1 &: \neg(((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)) \\ &\equiv \neg(\neg((\neg p \vee q) \wedge (\neg q \vee r)) \vee (\neg p \vee r)) \quad \text{Using } A \rightarrow B \equiv \neg A \vee B \\ &\equiv \neg\neg((\neg p \vee q) \wedge (\neg q \vee r)) \wedge (\neg\neg p \wedge \neg r) \quad \text{Using } \neg(A \vee B) \equiv \neg A \wedge \neg B \\ &\equiv ((\neg p \vee q) \wedge (\neg q \vee r)) \wedge (p \wedge \neg r) \quad \text{Using } \neg\neg A \equiv A\end{aligned}$$

We can now see that the resulting set of clauses becomes $\{\{\neg p, q\}, \{\neg q, r\}, \{p\}, \{\neg r\}\}$ and now we can start proving that $\neg F_1$ is unsatisfiable.

1. $\neg p, q$
2. $\neg q, r$
3. p
4. $\neg r$
5. q – Resolvent of 1. and 3.
6. r – Resolvent of 2. and 5.
7. $\square$ – Resolvent of 4. and 6.

Since we got an empty clause, $\neg F_1$ is unsatisfiable. Therefore, F_1 is valid.

$$F_2 : \forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z q(z)$$

To prove the validity of F_2 we do the exact same as we did with F_1 .

$$\begin{aligned}
\neg F_2 &: \neg(\forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z q(z)) \\
&\equiv \neg(\neg\forall x \exists y (p(x) \wedge (\neg p(y) \vee q(x))) \vee \forall z q(z)) && \text{Using } A \rightarrow B \equiv \neg A \vee B \\
&\equiv \neg\neg\forall x \exists y (p(x) \wedge (\neg p(y) \vee q(x))) \wedge \neg\forall z q(z) && \text{Using } \neg(A \vee B) \equiv \neg A \wedge \neg B \\
&\equiv \forall x \exists y (p(x) \wedge (\neg p(y) \vee q(x))) \wedge \neg\forall z q(z) && \text{Using } \neg\neg A \equiv A \\
&\equiv \forall x \exists y (p(x) \wedge (\neg p(y) \vee q(x))) \wedge \exists z \neg q(z) && \text{Using } \neg\forall x A \equiv \exists x \neg A \\
&\equiv \exists z \forall x \exists y ((p(x) \wedge (\neg p(y) \vee q(x))) \wedge \neg q(z)) && \text{Using } A \wedge \exists x B \equiv \exists x (A \wedge B) \\
&\equiv \forall x ((p(x) \wedge (\neg p(f(x)) \vee q(x))) \wedge \neg q(a)) && \text{Using Skolemisation}
\end{aligned}$$

The resulting set of clauses of $\neg F_2$ becomes $\{\{p(x)\}, \{\neg p(f(x)), q(x)\}, \{\neg q(a)\}\}$

1. $p(x_1)$
2. $\neg p(f(x_2)), q(x_2)$
3. $\neg q(a)$
4. $\neg p(f(a))$ – Resolvent of 2. and 3. with $\sigma = [x_2 \backslash a]$
5. $\square$ – Resolvent of 1. and 4. with $\tau = [x_1 \backslash f(a)]$

This proves the unsatisfiability of $\neg F_2$ which proves validity of F_2 .

Exercise O2.2

In order for the program to accept this new operator $\uparrow$ and be able to transform it into the clausal form we can tell it to transform all $\uparrow$ into something it can already handle from before. $A \uparrow B$ is equivalent to $\neg(A \wedge B)$ which is further equivalent to $\neg A \vee \neg B$ which it knows how to handle already.